

```
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
import cv2

from tensorflow.keras.models import Sequential, Model
from tensorflow.keras.layers import Conv2D, MaxPooling2D, AveragePooling2D
from tensorflow.keras.layers import Flatten, Dense, Input
```

```
image = cv2.imread("/content/fashion.jpg")
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
```

### Task 1

```
def show_image(image):
    plt.figure(figsize=(8,6))
    plt.imshow(image)
    plt.axis('off')
    plt.show()
show_image(image)
```



### Task 2

```
sample = image.astype("float32") / 255.0
resized_image = cv2.resize(sample, (28, 28))
# Convert to grayscale
resized_image = cv2.cvtColor(resized_image, cv2.COLOR_RGB2GRAY)
# Reshape to (1, 28, 28, 1) for the Conv2D layer
sample = resized_image.reshape(1, 28, 28, 1)

conv = Conv2D(
```

```

filters=8,
kernel_size=(3,3),
activation='relu'
)

feature_maps = conv(sample)
print(feature_maps.shape)

```

```
(1, 26, 26, 8)
```

```

fig, axes = plt.subplots(2,2, figsize=(10,5))

for i, ax in enumerate(axes.flat):
    ax.imshow(feature_maps[0,:,:,:i], cmap="gray")
    ax.set_title(f"Feature Map {i+1}")
    ax.axis("off")

plt.tight_layout()
plt.show()

```

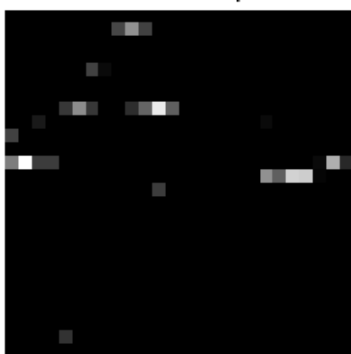
Feature Map 1



Feature Map 2



Feature Map 3



Feature Map 4



```

maxpool = MaxPooling2D(pool_size=(2,2))

max_output = maxpool(feature_maps)

print(max_output.shape)
avgpool = AveragePooling2D(pool_size=(2,2))

avg_output = avgpool(feature_maps)

print(avg_output.shape)

```

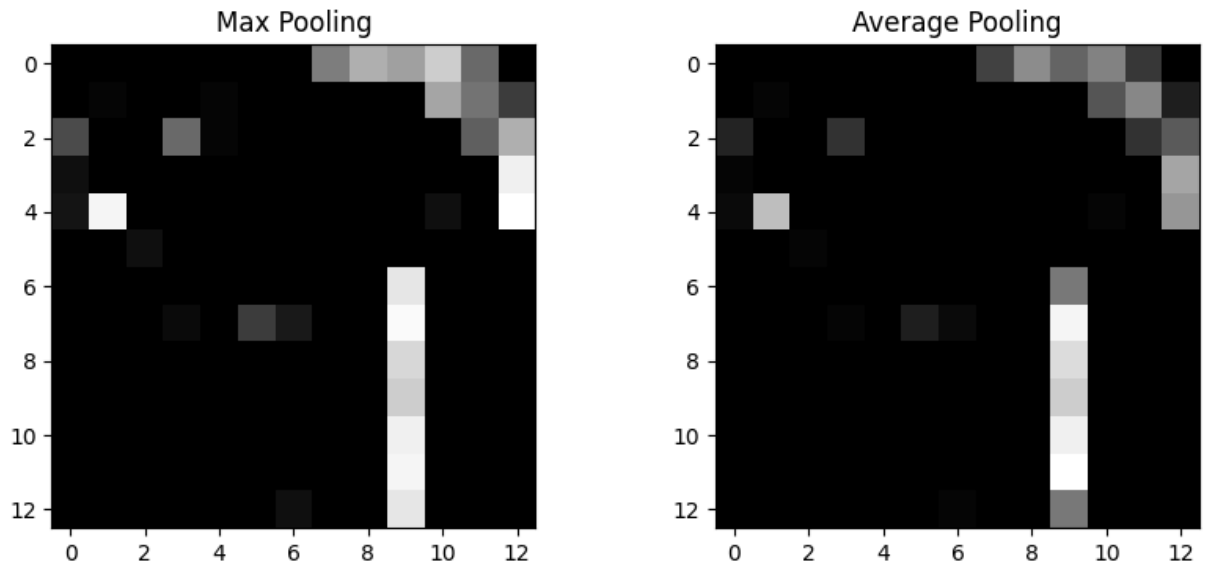
```
(1, 13, 13, 8)
(1, 13, 13, 8)
```

```
plt.figure(figsize=(10,4))

plt.subplot(1,2,1)
plt.imshow(max_output[0,:,:], cmap="gray")
plt.title("Max Pooling")

plt.subplot(1,2,2)
plt.imshow(avg_output[0,:,:], cmap="gray")
plt.title("Average Pooling")

plt.show()
```



```
model = Sequential([
    Conv2D(32,(3,3),activation='relu',input_shape=(28,28,1)),
    MaxPooling2D(2,2),
    Conv2D(64,(3,3),activation='relu'),
    MaxPooling2D(2,2),
    Flatten(),
    Dense(128,activation='relu'),
    Dense(10,activation='softmax')
])
model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)
model.summary()
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning:
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Model: "sequential"

| Layer (type)                   | Output Shape       | Param # |
|--------------------------------|--------------------|---------|
| conv2d_1 (Conv2D)              | (None, 26, 26, 32) | 320     |
| max_pooling2d_1 (MaxPooling2D) | (None, 13, 13, 32) | 0       |
| conv2d_2 (Conv2D)              | (None, 11, 11, 64) | 18,496  |
| max_pooling2d_2 (MaxPooling2D) | (None, 5, 5, 64)   | 0       |
| flatten (Flatten)              | (None, 1600)       | 0       |
| dense (Dense)                  | (None, 128)        | 204,928 |
| dense_1 (Dense)                | (None, 10)         | 1,290   |

Total params: 225,034 (879.04 KB)

Trainable params: 225,034 (879.04 KB)

Non-trainable params: 0 (0.00 B)